

EXPNO:3	CLASSIFICATION WITH DECISION TREES
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AIM

To implement a **Decision Tree classifier** and evaluate its performance using **accuracy score** and **confusion matrix** on a real-world dataset.

ALGORITHM

1. Import the necessary libraries.
2. Load a classification dataset such as the **Iris dataset**.
3. Separate the features and target variable.
4. Split the dataset into training and testing sets.
5. Preprocess the data if required.
6. Create a DecisionTreeClassifier using sklearn.tree.
7. Train the Decision Tree model using the training data.
8. Predict the target values using the test data.
9. Calculate the **accuracy score**.
10. Generate the **confusion matrix**.
11. Visualize the Decision Tree.
12. Display the results.

CODE:

```
from sklearn.datasets import load_iris

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.model_selection import train_test_split

from sklearn.metrics import confusion_matrix, accuracy_score

import matplotlib.pyplot as plt

import seaborn as sns

iris = load_iris()
```

```
X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

dt_model = DecisionTreeClassifier(
    criterion='gini',
    random_state=0
)

dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

acc = accuracy_score(y_test, y_pred)

print("Confusion Matrix:\n", cm)

print("Accuracy Score:", acc)

plt.figure(figsize=(6, 5))

sns.heatmap(
    cm,
    annot=True,
    cmap="Blues",
    fmt="d",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)
```

```
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

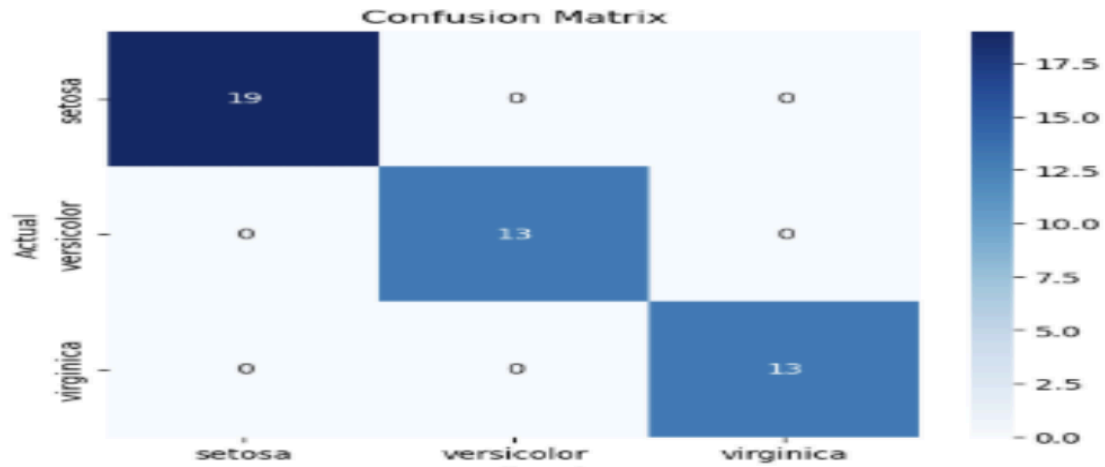
plt.figure(figsize=(12, 8))

plot_tree(
    dt_model,
    filled=True,
    feature_names=iris.feature_names,
    class_names=iris.target_names
)

plt.title("Decision Tree Visualization")
plt.show()
```

OUTPUT:

Confusion Matrix:
[[19 0 0]
[0 13 0]
[0 0 13]]
Accuracy Score: 1.0



Decision Tree Visualization

